Optogenetic Novel Tools: Cytoskeleton-Dependent Regulation of Endoplasmic Reticulum Dynamics Revealed by Intracellular Spatiotemporal Profiling and Quantitative Live-Cell Imaging

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Introduction

In live cells, the cytoskeleton and the endoplasmic reticulum (ER) network work as ferries to transport cargo along cytoskeletal highways, powered by molecular motors. The disruption of the transportation system will cause cell malfunction and pathological phenomena. An effective way of monitoring the trajectories of their movements is yet to be explored. Therefore, this unprecedented research applies a methodology derived from optogenetics to resolve this problem on ER dynamics and its relationship with the cytoskeleton.

By equipping the photosensitive proteins CRY2 that respond to blue lasers, we can steer organelle activities with pinpoint precision. It can trigger the homo-interaction of the specific intracellular domain in the ER membrane, showing many red fluorescent 'punctum', to successfully visualize the ER membrane movement under high-resolution live-cell imaging. Then, using drug disruption can selectively block different cytoskeletons, including microtubules, actin filaments, with nocodazole, and latrunculin B, to simulate the pathological situations and pinpoint which cytoskeletal routes are pivotal to ER dynamics.

This novel tool will reveal how the cytoskeleton works with the ER and will shape the intracellular activities. It can provide researchers with a novel way to explore ER and cytoskeletons' traffic in health and disease by quantifying the effects of puncta's movement.

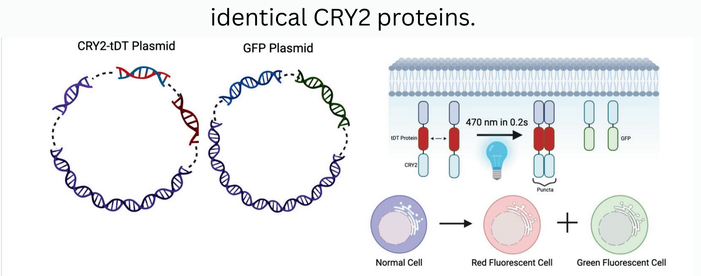
Hypothesis & Aims

Hypothesis

It is hypothesised that the drug disruption of active filaments and microtubules will seriously affect the ER membrane structure, and the movement of puncta will stall or be disrupted everywhere.

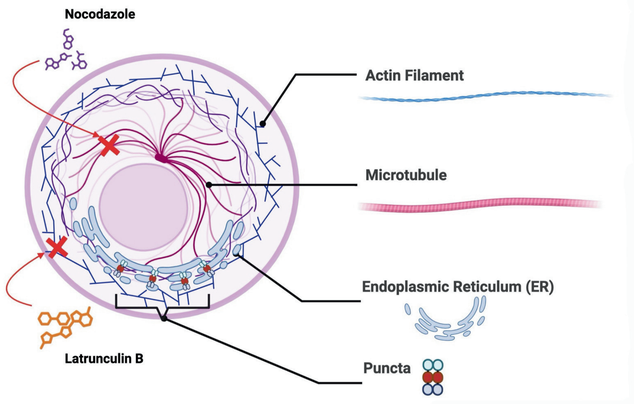
Aim 1

To validate the expression of fluorescent protein: tDT and GFP on the ER membrane, triggered by CRY2 and the formation of puncta due to homo-interaction happened between two identical CRY2 proteins.



Aims 2:

To investigate the roles of different cytoskeleton, including active filaments, microtubules, that can affect the speeds and efficiencies of movement of ER through adding disruptive drugs to stimulate the pathological state of the cell, and collect reelvent data to compare with normal cells.



Discussion

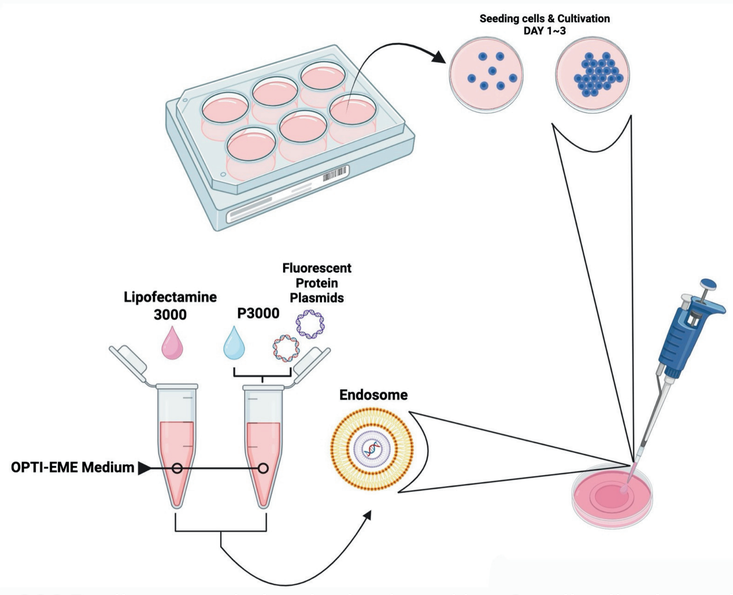
The successful formation of puncta through homo-interaction between puncta to verify that this strategy remains both stable and reliable in other living cells. These results showed red and green fluorescence on the ER membrane and puncta. Five sets of data from three cell groups were compared on Fiji software, showing that the delivery speed shrank in nocodazole group and observed vigorous puncta movement, whereas the confinement ratio and duration of them increased when compared with the normal group and Latrunculin B group. This shows that drugs have imposed perturbation on the cytoskeleton, causing negative effects on the ER when it contracts toward the nucleus.

However, A rigorous quantitative framework for classifying puncta motility is still lacking, partly because certain puncta are transient and may depend on other cytoskeleton-ER interactions, particularly intermediate-filament contacts, for which no selective pharmacological disruptor exists. To address these gaps, we propose a method of siRNA-mediated knockdown of genes encoding intermediate-filament components to investigate it in future research.

Overall, such an application will help establish the CRY2 photosensitive trigger as a robust tool for intracellular biomedical research and push the boundaries of cellular-activity research.

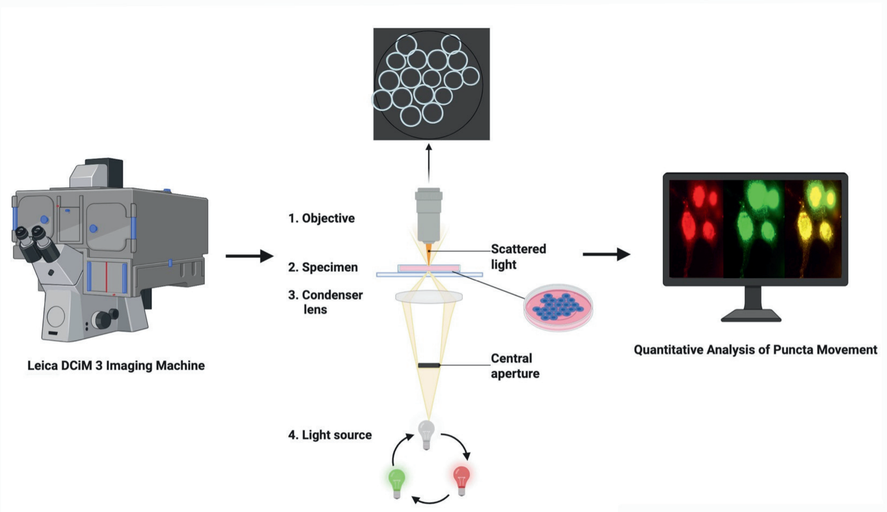
Materials & Methods

Step 1: Cell Subculturing & Transfection



COS-7 cells are seeded and subcultured in a 6-well cell culture plate. Two plasmids, containing sequences of CRY2 and two fluorescent proteins, are diluted in a mixture of commercial reagents and added to COS-7 cells on a confocal dish.

Step 2: Life-cell imaging & ER Dynamic Quantification



COS-7 Cells are imaged on an alive-cell confocal microscope. Two fluorescence channels, FC1 (red light) and FC2 (green light), are set as follows, with the blue pulse to maintain CRY2 activation. Two disruptive drugs (nocodazole and latrunculin B) to stall the cytoskeleton's function are applied for comparison. The puncta movement is measured by Fiji software for quantitaive analysis.

Results

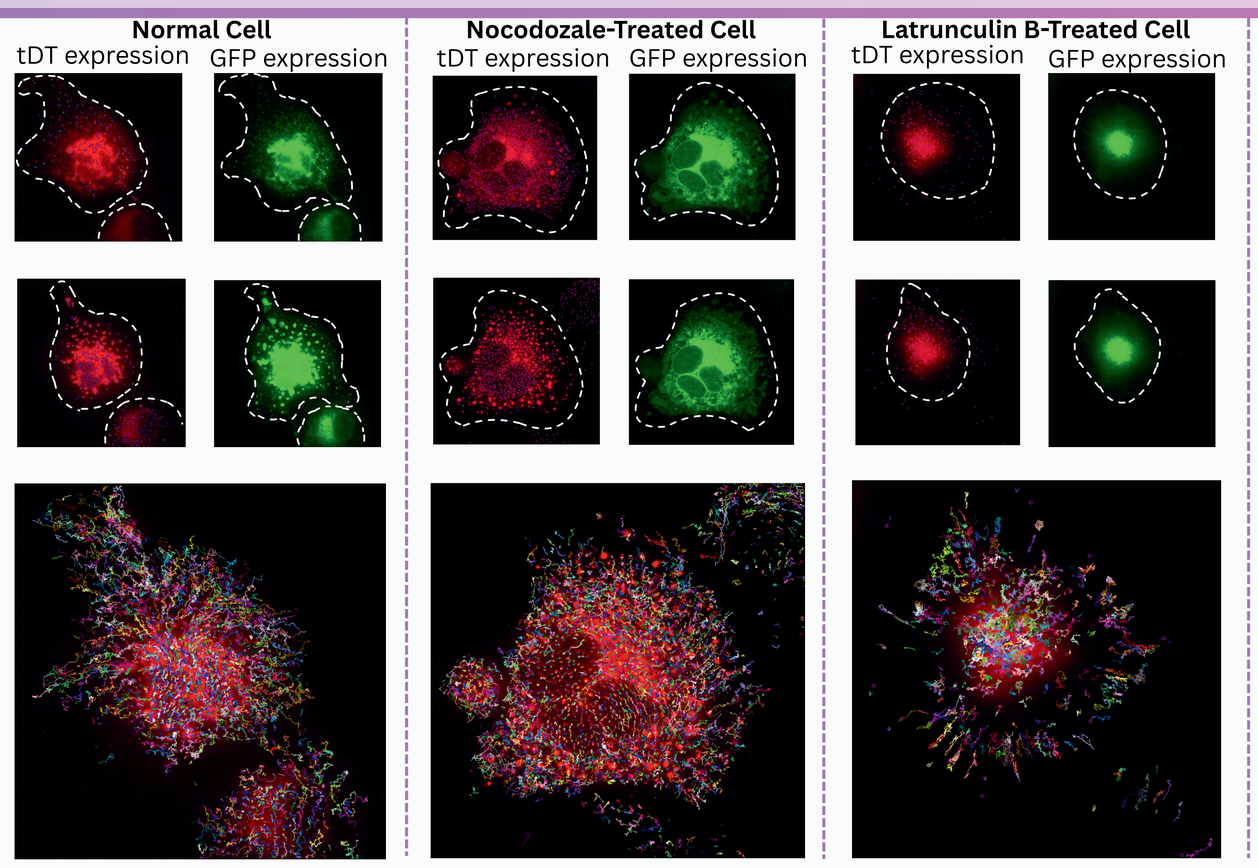


Fig.1: The imaging recording for cell representatives. From left to right are the normal group, nocodazole and latrunculin B. Cells show red and green fluorescence after being illuminated by lights with 470 nm and 550 nm wavelengths separately. The CRY2 proteins are triggered to form puncta that are tracked, showing as colourful trajectories

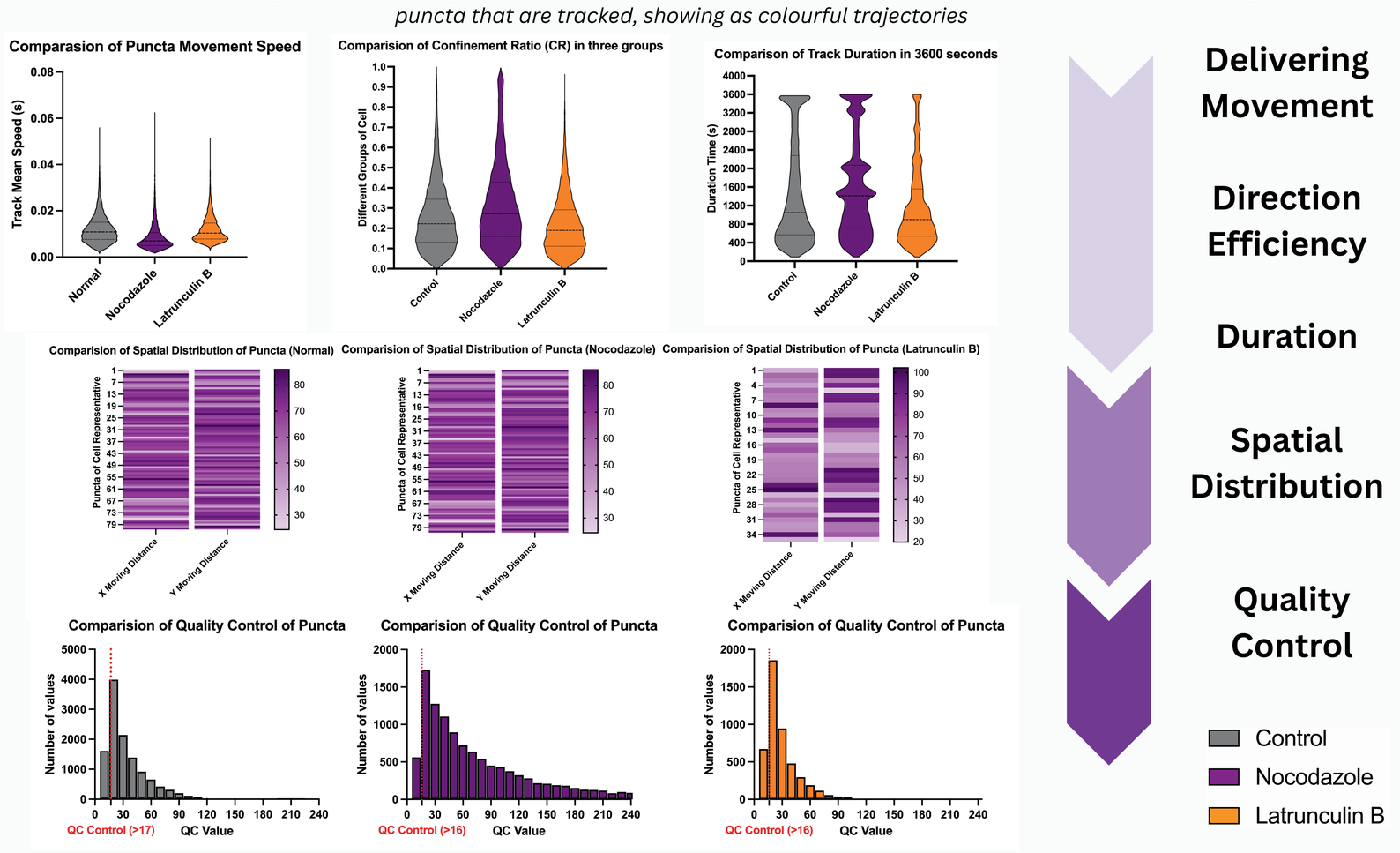


Fig.2: The tracking data comparisons of puncta movement in three groups of cells (one normal, one nozocodale, one latrunculin B)

References & Acknowledgment

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